Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2019-II-Q02] | Part A (Q1-Q5)

Answer: 107

1. Let u_k be the visit probability. The last step to k comes uniquely from $k-1$ by one or $k-2$ by two, so $u_k = (u_{k-1} + u_{k-2})/2$, with $u_1 = 1, u_2 = 1/2$.

2. Iteration gives $u_3 = 3/4, u_4 = 5/8, u_5 = 11/16, u_6 = 21/32, u_7 = 43/64$. Thus $p + q = 107$.

Source / 来源

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2023-I-Q06] | Part B (Q6-Q10)

Answer: 051

1. The prediction cannot affect the remaining deck. Therefore maximizing each immediate success probability is optimal: always choose a color with the larger remaining count.

2. For $E(r,b)$, the expected future successes, use $E(r,0)=r, E(0,b)=b$ and $E(r,b)=[\max(r,b)+rE(r-1,b)+bE(r,b-1)]/(r+b)$.

3. By symmetry, the necessary values are $E(1,1)=3/2, E(2,1)=7/3, E(2,2)=17/6, E(3,1)=13/4, E(3,2)=18/5$.

4. Thus $E(3,3)=1/2+18/5=41/10$, and $m+n=51$.

Source / 来源

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2026-II-Q07] | Part B (Q6-Q10)

Answer: 754

1. Condition on exactly k Alice/Bob rolls before the first Carol roll. For $k \geq 4$, the good binary strings number $2^k - 2(k+1)$, since strings with at most one occurrence of either person are disjoint bad families.

2. Each such length-k string followed by Carol has probability $3^{-(k+1)}$. Thus $P = \frac{1}{3}\sum_{k\geq 4}[2^k - 2(k+1)]/3^k$. The two sums are $16/27$ and $11/54$, using geometric-series differentiation.

3. Hence $P = \frac{1}{3}(16/27 - 11/54) = 7/54$ and $100m + n = 754$.

Source / 来源

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2015-I-Q12] | Part C (Q11-Q15)

Answer: 431

1. Write each chosen subset in increasing order. The unchosen integers form 1001 gaps: before the first chosen number, between consecutive chosen numbers, and after the last. Their sizes are nonnegative and sum to 1015.

2. The correspondence between subsets and these gap vectors is a bijection, so all weak compositions are equally likely. Permuting gap coordinates preserves their distribution, giving mean first gap $1015/1001$.

3. The smallest chosen number is one plus that gap. Its mean is $1 + 1015/1001 = 2016/1001 = 288/143$. Thus $p + q = 431$.

Source / 来源

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2016-I-Q13] | Part C (Q11-Q15)

Answer: 273

1. Only height matters. Let T_j be the expected remaining jumps at height j. First-step analysis gives $T_{24} = 0$, $T_0 = 3 + T_1$, and $T_{j+1} - 2T_j + T_{j-1} = -4$ for $1 \leq j \leq 23$.

2. Write differences $d_j = T_j - T_{j-1}$. Then $d_1 = -3$ and $d_{j+1} = d_j - 4$, so $d_j = 1 - 4j$.

3. Thus $T_{21} = -\sum_{j=22}^{24} d_j = (87 + 91 + 95) = 273$. Finite expectation follows, for example, because every block of24 jumps has a fixed positive chance of all upward choices and hence absorption.

Source / 来源

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2016-II-Q13] | Part C (Q11-Q15)

Answer: 371

1. There are $6! = 720$ configurations. Every score is at least2 and at most7, the latter because the six occupied values sum to42. Therefore the total score is $2 \cdot 720 + \sum_{j=3}^{7} B_j$, where B_j counts scores at least j.

2. To have score at least j, row r must use column $c \geq j - r$. For the first $j - 2$ rows, allowed sets are nested, giving exactly $8 - j$ unused choices at each step; the remaining $8 - j$ rows can be permuted freely. Hence $B_j = (8 - j)^{j-2}(8 - j)!$.

3. The five tail counts are $600, 384, 162, 32, 1$. Mean score is $(1440 + 600 + 384 + 162 + 32 + 1)/720 = 2619/720 = 291/80$, so $p + q = 371$.

Source / 来源

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2018-II-Q13] | Part C (Q11-Q15)

Answer: 647

1. Use prefix states zero (no match), one (last result one), and two (last two results one-two). Let u_j be the expected value of $(-1)^R$ for the number R of remaining rolls from each state. Stopping is almost sure, since independent three-roll blocks have a fixed positive chance of being one-two-three.

2. Each next roll changes parity, so conditioning gives $6u_0 = -u_1 - 5u_0$, $6u_1 = -u_1 - u_2 - 4u_0$, and $6u_2 = -1 - u_1 - 4u_0$. The constant term corresponds to immediate completion from state two.

3. The first two yield $u_1 = -11u_0, u_2 = 73u_0$. The third gives $431u_0 = -1$, so the odd-roll probability is $(1 - u_0)/2 = 216/431$.

4. The fraction is reduced and $m + n = 647$.

Source / 来源

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2021-I-Q12] | Part C (Q11-Q15)

Answer: 019

1. The three cyclic gaps stay even. Before collision their unordered possibilities are $A = (4,4,4)$, $B = (2,4,6)$, $C = (2,2,8)$. Enumerating the eight jump-sign triples gives transition counts: $A \to (A,B) = (2,6)$; $B \to (A,B,C,\text{stop}) = (1,4,1,2)$; $C \to (B,C,\text{stop}) = (2,2,4)$.

2. With expected remaining times a, b, c, first-step equations are $a = 1 + a/4 + 3b/4$, $b = 1 + a/8 + b/2 + c/8$, $c = 1 + b/4 + c/4$.

3. The last gives $c = 4/3 + b/3$ and the first $a = 4/3 + b$. Substitution gives $b = 4$, hence $a = 16/3$ and the requested sum 19.

Source / 来源

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2025-I-Q13] | Part C (Q11-Q15)

Answer: 204

1. With probability one, all interior crossings are distinct except the prescribed diameter crossing. Region count equals $1 + 27+$ the number of crossing pairs. The diameters contribute one crossing, and each random chord crosses each diameter with probability $4/6 = 2/3$.

2. For two random chords there are 36 equally likely ordered quadrant-pair choices. Two combinations force crossing, four forbid it, and the remaining thirty cross with probability $1/2$ by endpoint-order symmetry. Hence the crossing probability is $17/36$.

3. Linearity of expectation gives $28 + 1 + 25 \cdot 4/3 + \binom{25}{2} \cdot 17/36 = 204$.

Source / 来源